

Fluorescence Imaging in Veterinary Medicine



Fluorescence Imaging

Fluorescence imaging is becoming increasingly popular as a visual tool in surgery as clinical studies in animals pave the way for applications in human medicine. KARL STORZ thus offers Opal1® technologies for near infrared (NIR/ICG) and Blue Light Imaging (BLI). Each technology also holds great potential for enhanced diagnosis and treatment of disease in veterinary patients.

Opal1® Technologies

In 1852, George Gabriel Stokes described the mineral Fluorite as emitting blue light following exposure to UV-light. He termed the phenomenon as “Fluorescence” and substances with this property as “Fluorophores”.

The ability to emit fluorescence is very common in nature. The sensitivity of delocalized electrons in aromatic ring structures is responsible for this. The light energy excites the delocalized electrons into a higher state of energy. When the electrons return to the ground state, the light energy is emitted as fluorescence. The emitted fluorescence is lower in energy as the exciting light energy is partly lost as heat.

Aromatic ring structures are the main components of biological substances such as DNA, proteins and sugars. Since the 1960’s their ability to emit fluorescence has been used for fluorescence imaging in life sciences and in medicine.

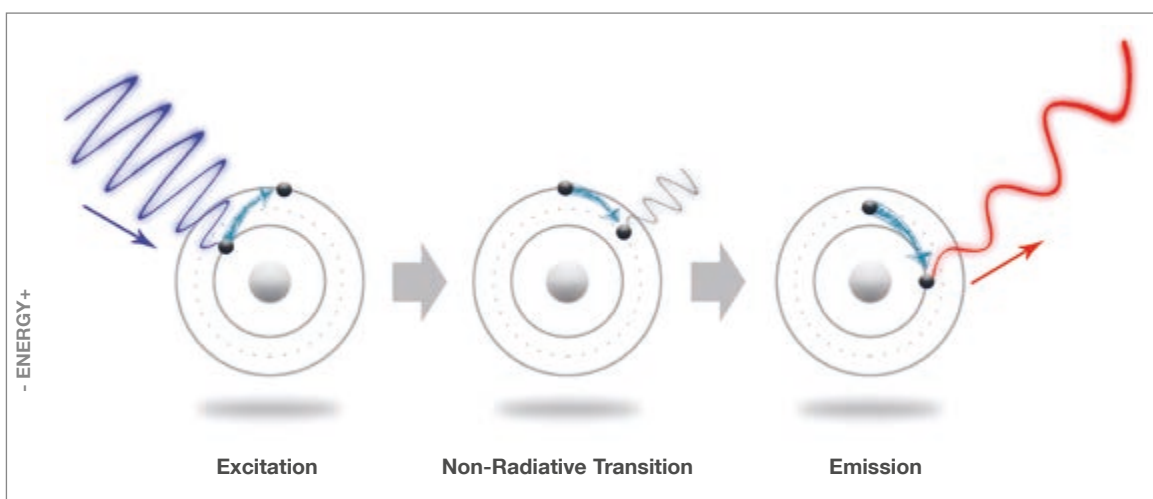


Fig. 1: Principle of fluorescence/Stokes shift

NIR/ICG with Rubina®

The oldest known approved near-infrared (NIR) fluorescent dye in medicine, ICG, has been discovered as a tool to visualize the anatomy and the lymphatic system as well as to detect tumors and to assess perfusion. Therefore, a wide range of applications are possible.

ICG is a drug which has been approved for use in humans by the FDA since 1959 for cardiac and liver function testing. The tricarbon dye has an excitation maximum of $\lambda_{Em} = 805\text{ nm}$ and an emission maximum of $\lambda_{Ex} = 835\text{ nm}$.

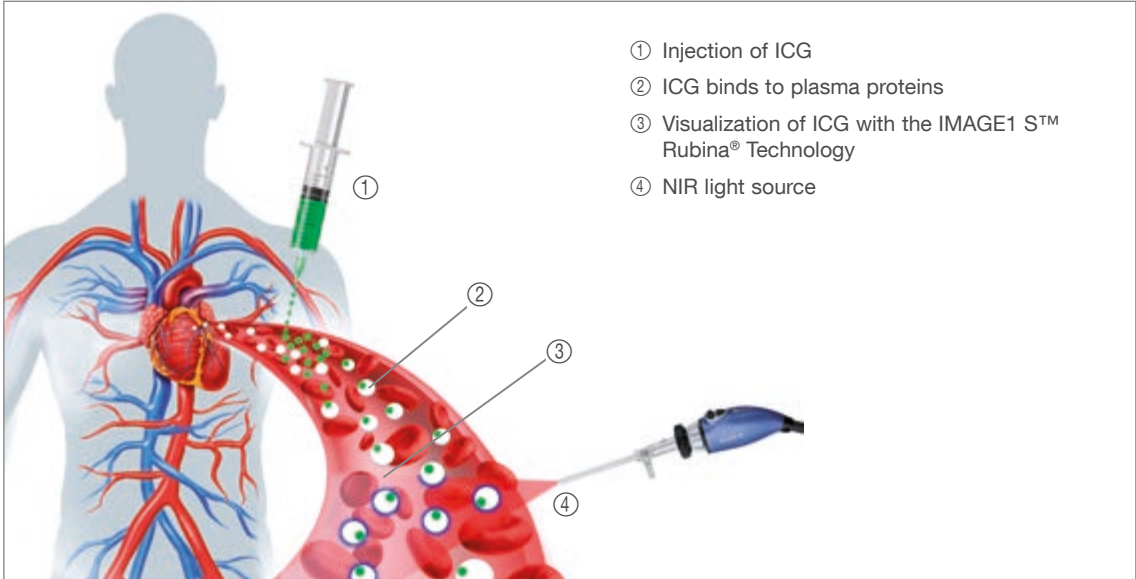


Fig. 2: Principle of ICG

For example, ICG can be administered intravenously where it binds to plasma proteins (albumin), thereby remaining in the bloodstream due to size exclusion. From the bloodstream, ICG is transported to the liver, where it is excreted via the bile into the duodenum. This exclusive excretion into bile makes it an ideal tool for the delineation of the extra-hepatic biliary tree.



White light



ICG Enhanced Fluorescence
Guided Laparoscopic Surgery

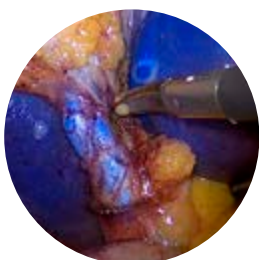
NIR/ICG Visualization Modes

The Rubina® components offer users various new modes for visualizing the NIR/ICG signal. This includes the overlay of NIR/ICG data onto the standard white light image or alternatively the monochromatic visualization of the infrared signal alone.



Overlay

In Overlay mode, the regular white light image is combined with the NIR/ICG data to generate an overlay image.



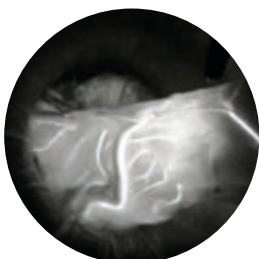
Green or blue – it's up to you

Depending on your preferences and application, the NIR/ICG data can be displayed as a green or blue overlay.



Intensity Map

Displays the intensity of the NIR/ICG signal using a color scale in an overlay image.



Monochromatic

In this mode, the NIR/ICG signal alone is displayed in white on a black background to achieve the greatest possible differentiation.

Indications in Veterinary Patients

The IMAGE1 S™ Rubina® technology has already been used for several different indications in veterinary medicine. The four main indications are:

- Blood perfusion assessment
- Lymphatic mapping
- Visualization of anatomical structures (e.g., bile duct imaging)
- Tumor identification

The following list is a selection of the veterinary applications of the NIR/ICG technology to date:

- Identification of vascular integrity
- Lymphatic mapping for oncology
- Evaluation of lymphatic disease
(chylothorax, lymphangiectasia, assessment of lymphedema)
- Biliary function and anatomy (obstruction, leakage)
- Tumor margin localization
- Identification of metastasis (primarily liver)
- Lung segmentectomy
- Ureteral identification
- Skin flap/graft viability

Example Applications of NIR/ICG



Visualization of Hepatocellular Carcinoma and Metastases



Bile Visualization



Visualization of Hepatocellular Carcinoma and Metastases

Opal1® Technology for NIR/ICG



- ① **IMAGE1 S™ camera system**
 - 4K or FULL HD image quality
 - NIR/ICG modes + S-Technologies for image enhancement
- ② **NIR/ICG telescope**
 - Telescopes for optimal fluorescence excitation & detection, which can be used for white light and NIR/ICG applications
 - Standard endoscopes can also be used with the NIR snap-on filter
- ③ **Camera head options**
 - 3-chip FULL HD camera head with high resolution and optimal NIR light sensitivity
 - 4K camera head with pristine image quality in both white light and NIR/ICG modes
 - 3D videoscope delivers 3D technology with 4K image quality
- ④ **POWER LED Rubina®**
 - Excitation of ICG and autofluorescence in the near infrared range
 - Laser-free LED light source
 - Intuitive touch screen interface
- ⑤ **Autoclavable fiber optic light cable**
 - Optimal light transmission in the white light and NIR spectral range
 - Autoclavable
- ⑥ **Footswitch**
 - Fast switch between white light and fluorescence mode

Literature, see pages 13 and 14

IMAGE1 S™ Saphira™ Technology for Blue Light Imaging (BLI)

Fluorescence imaging with 5-aminolevulinic acid (5-ALA)

The identification and total removal of malignant tissue determines the life expectancy of a cancer patient. Many researchers try to overcome this problem by developing specific tumor markers. One of the most promising substance groups are the protoporphyrin IX-producing (PPIX) drugs. 5-aminolevulinic acid (5-ALA) and its approved derivative Hexvix® from Photocure (Cysview® in USA) and Gliolan® from Medac are precursors of PPIX in the heme biosynthesis. The substances are internalized and metabolized by all cells in the body. Malignant cells have a metabolic defect, which results in the accumulation of PPIX in the cells. Since PPIX has fluorescent properties, malignant tissue can be visualized by enhancing the PPIX fluorescence (PPIX absorption maximum $\lambda_{Em} = 420\text{ nm}$, PPIX fluorescence emission maximum $\lambda_{Ex} = 630\text{ nm}$).

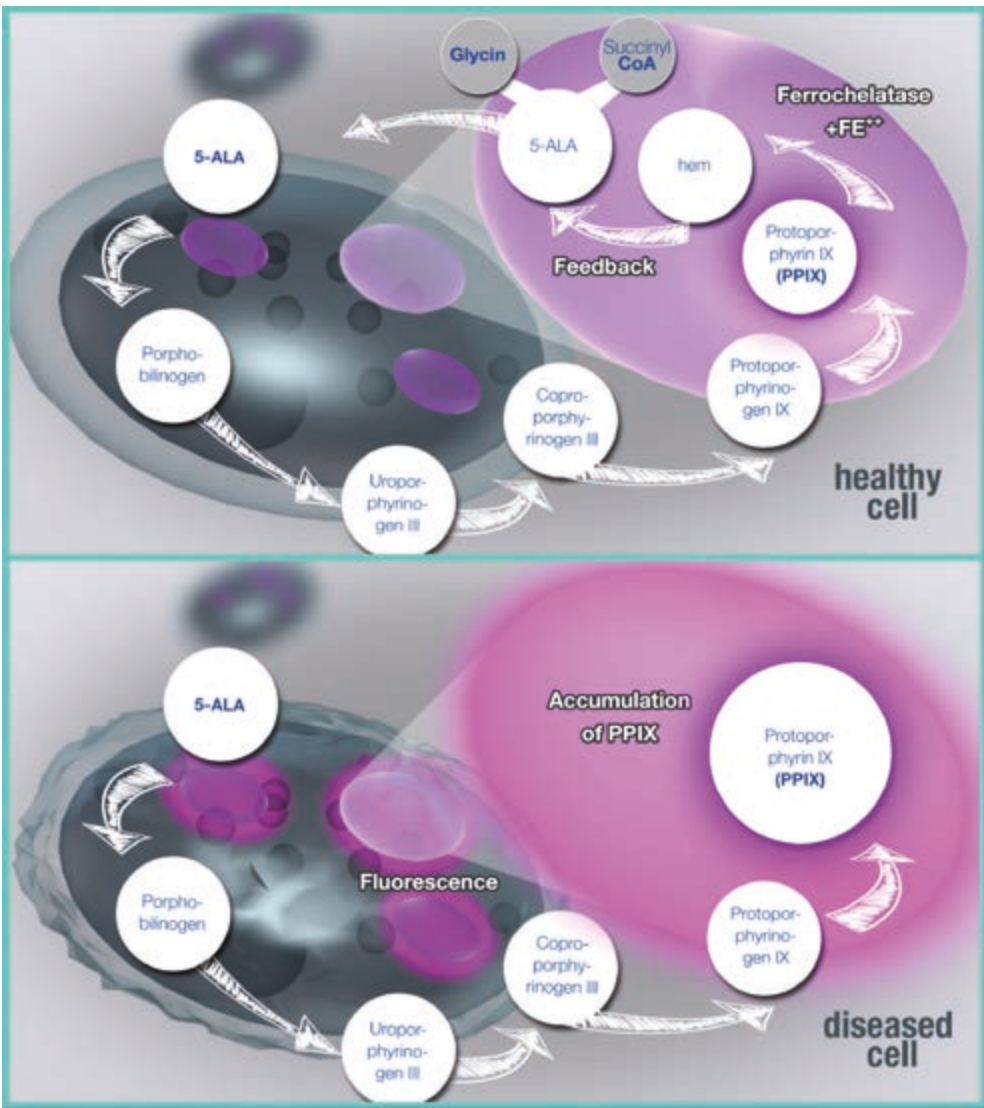
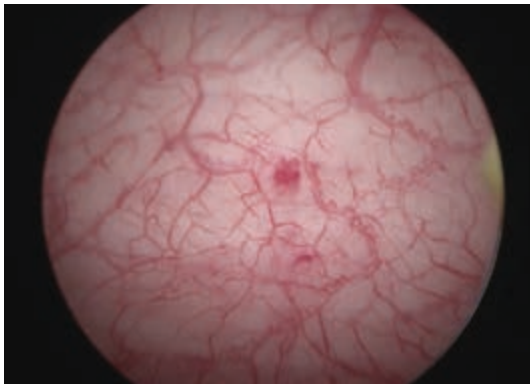


Fig. 7: 5-ALA, Hexvix® or Gliolan® are converted into precursors of PPIX in the heme biosynthesis.

Indications

Two primary indications for blue light imaging (BLI) with PPIX precursors are known: 5-ALA or Gliolan® is used for glioma identification in neurosurgery. Hexvix® (Cysview® in USA) is distilled into the bladder for bladder cancer detection (Fig. 8).

➤ Demarcation of Bladder Tumors via blue light imaging



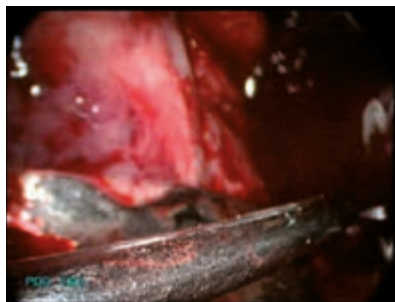
A) White light



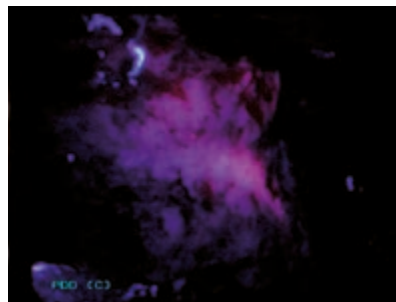
B) Blue light imaging with Saphira™ technology based on IMAGE1 S™³

Fig. 8: Fluorescence diagnosis of a bladder tumor with Hexvix®

➤ Visualization of Glioblastoma via Fluorescence



A) White light



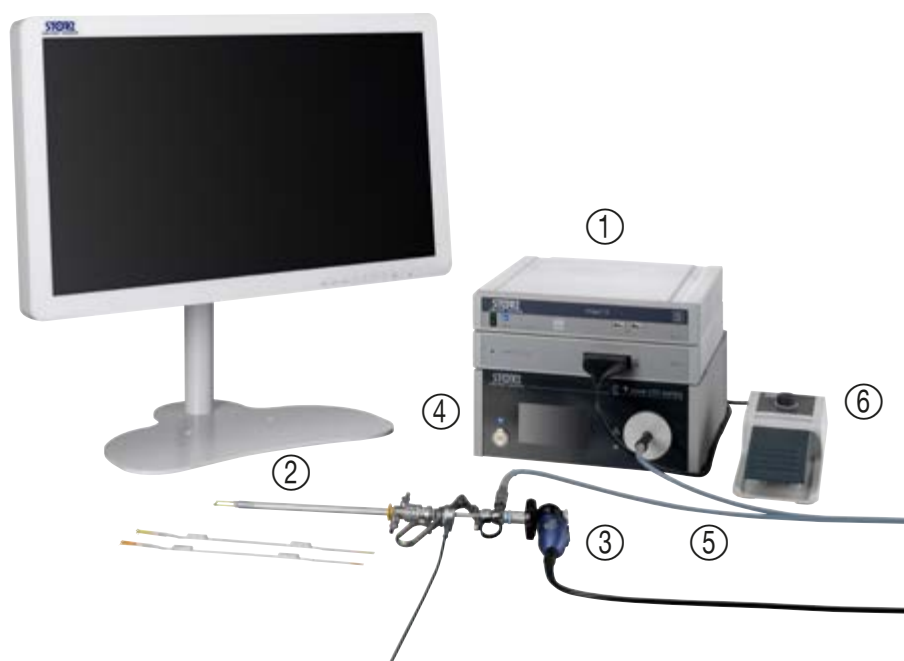
B) PPIX-accumulating glioblastoma (red)⁴

Fig. 9: Fluorescence diagnosis of a glioblastoma with Gliolan®

³ Images courtesy of PD Dr. med. Carsten Kempkensteffen, Charité University Medical Center Berlin, Germany

⁴ Images courtesy of Prof. Potapov, Burdenkow Neurosurgical Institute Moscow, Russia

IMAGE1 S™ Saphira™ Technology for Blue Light Imaging



① **IMAGE1 S™ camera system**

- BLI mode with different intensity levels and SPECTRA A and SPECTRA B in white light lead to optimal illumination and contrast enhancement in white light only with a display in the cyan spectral range

② **BLI telescope**

- Telescopes for optimal fluorescence excitation & detection, which can be used for white light and BLI applications

③ **Camera head**

- 1-chip FULL HD HX camera head with high resolution and optimal BLI light sensitivity
- Standard and pendulum camera heads available
- Standard endoscopes can also be used with a snap-on filter

④ **POWER LED Saphira™**

- LED based light source with white and blue light
- Constant light intensity
- Intuitive touch screen interface

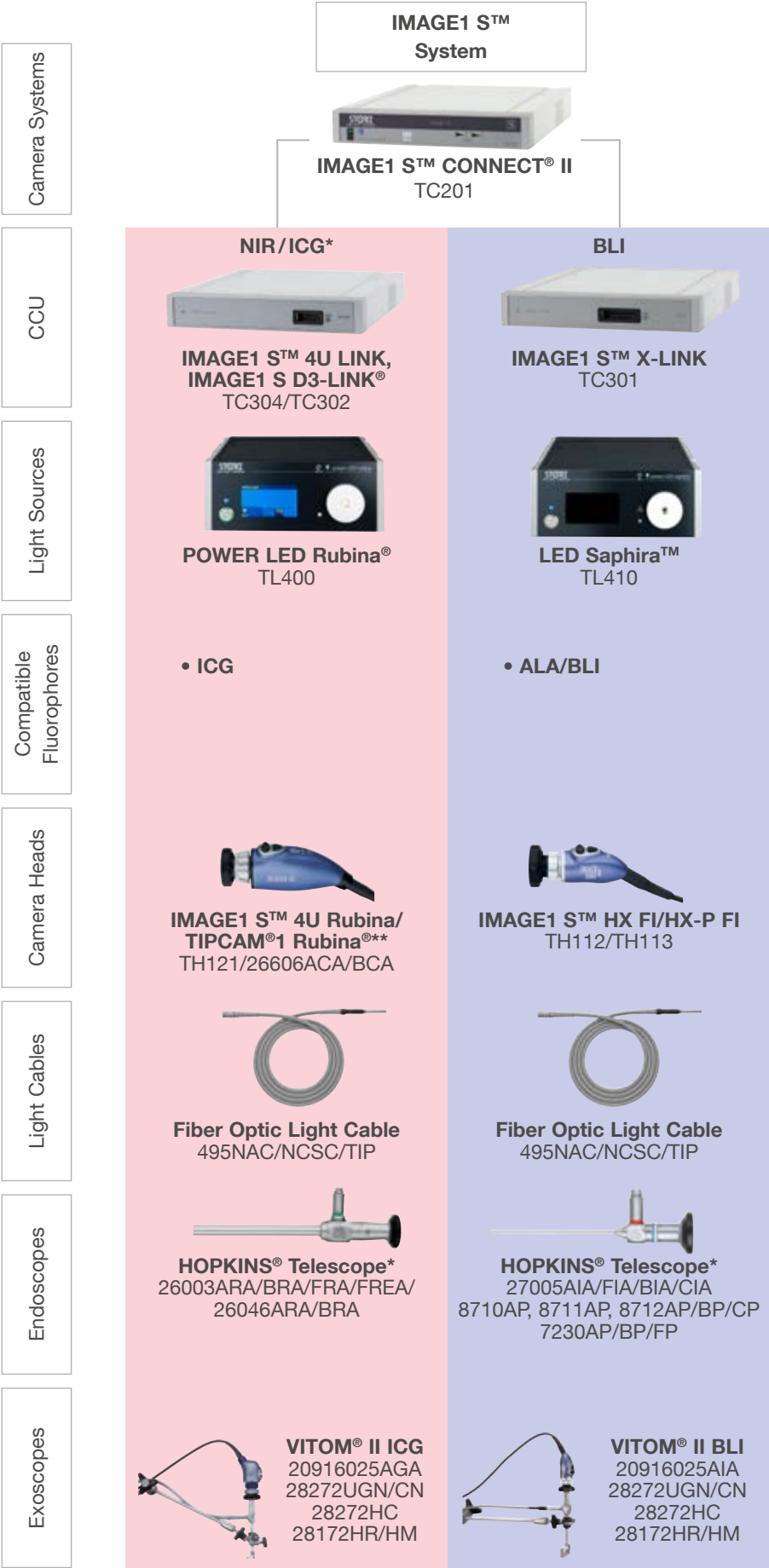
⑤ **Fluid light cable**

- Optimal light transmission of white light and fluorescence

⑥ **Footswitch**

- Fast switch between white light and fluorescence mode

Literature, see page 15



* Standard KARL STORZ telescopes can also be utilized when used in conjunction with a compatible snap-on filter (see page 12)
** 3D function requires use of TIPCAM® videoscopes, no independent camera head required. TIPCAM®1 Rubina® is only suitable for use with ICG.

Snap-on Filters

for use with standard KARL STORZ telescopes



60100040 **Snap-on Filter for ICG, for use with IMAGE1 S™ POWER LED Rubina®**

20100034 **Snap-on Filter for 5-ALA/BLI, for use with IMAGE1 S™ Saphira™**

Please note:

If you have any questions or require further information regarding fluorescence in veterinary medicine, please contact VET@KARLSTORZ.com.

Check out our schedule of upcoming hands-on training courses at
<http://go.karlstorz.com/eventsVET>



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